

Department of Artificial Intelligence and Machine Learning

QUESTION BANK FOR CIE 1

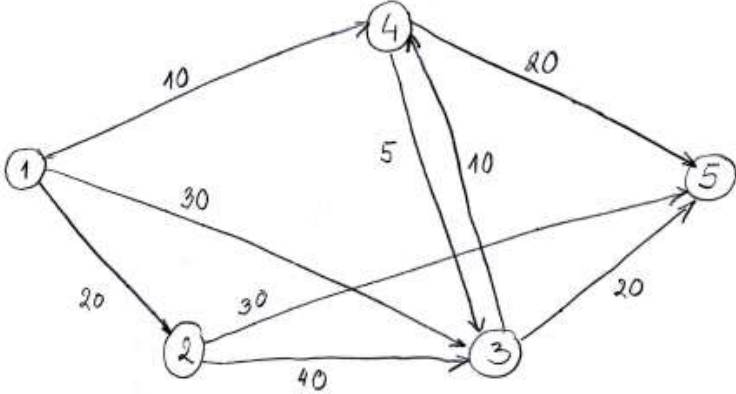
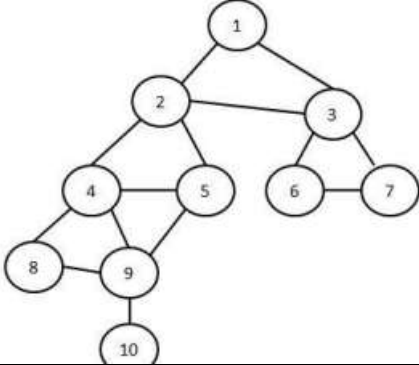
IV Sem (Term: Feb-June 2026)

Algorithms Laboratory (AIL47/ADL47)

Note:

- 1) For the questions in (a) that is related to asymptotic bounds program should be written and for the rest algorithm should be written.
- 2) Programs can be written or implemented using C / C++/ Java programming language.
- 3) The built-in modules should not be used for implementation (except time module and random module)

Sl.No	Questions	CO Mapping																								
1.	a) Given $f(n)= 7n+5$, Write a program to prove that $f(n)= O(n)$ and find the n_0 value. Plot a graph for $f(n)$ and $c*g(n)$ where c is a constant and for varying n values (10 to 30). b) Given a set of men and women design and implement Gale- Shapeley algorithm to determine the stable set of marriages among them. Comment on the efficiency and the time complexity of the same. Assumptions: Men propose first according to their preference list. Women can choose a better partner based on the preference. Men's preference list <table><tr><td>A</td><td>V</td><td>W</td><td>X</td></tr><tr><td>B</td><td>W</td><td>V</td><td>X</td></tr><tr><td>C</td><td>V</td><td>W</td><td>X</td></tr></table> Women's preference list <table><tr><td>V</td><td>A</td><td>B</td><td>C</td></tr><tr><td>W</td><td>B</td><td>C</td><td>A</td></tr><tr><td>X</td><td>C</td><td>A</td><td>B</td></tr></table>	A	V	W	X	B	W	V	X	C	V	W	X	V	A	B	C	W	B	C	A	X	C	A	B	CO1
A	V	W	X																							
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V	A	B	C																							
W	B	C	A																							
X	C	A	B																							
2.	a) Given $f(n)= 3n^2+4n+3$, Write a program to prove that $f(n)= \Omega(n)$ and find the n_0 value. Plot a graph for $f(n)$ and $c*g(n)$ where c is a constant and for varying n values (10 to 30). b) A GPS navigation system needs an approach to discover the reachable areas in a given geographical region from a given source area. Design and implement an algorithm to find which nodes can be reached from a given source node for the following graph. Comment on the efficiency and the time complexity of the same.	CO1																								

		
3.	a) Given $f(n) = 7n^2 + 7n + 5$, Write a program to prove that $f(n) = \theta(n^2)$ and find the n_0 value. Plot a graph for $f(n)$, $c_1 * g(n)$ and $c_2 * g(n)$ where c_1, c_2 is a constant and for varying n values (10 to 30). b) Design and implement merge sort algorithm that takes random number input and displays the execution time required. State the design strategy used and time complexity of the same.	CO2
4.	a) Given $f(n) = 4n + 3$, Write a program to prove that $f(n) = O(n)$ and find the n_0 value. Plot a graph for $f(n)$ and $c * g(n)$ where c is a constant and for varying n values (10 to 30). b) Three users in an online music portal listen to a playlist of 8 songs that are numbered from 1 to 8 in a random order. Each user needs to be recommended to another user playlist's order that has minimum number of inversions. Design and implement an algorithm to determine the number of inversions. State the design strategy used and time complexity of the same.	CO2
5.	a) Given $f(n) = 2n + 3n + 5$, Write a program to prove that $f(n) = \Omega(n)$ and find the n_0 value. Plot a graph for $f(n)$ and $c * g(n)$ where c is a constant and for varying n values (10 to 30). b) A person starts from a given source and reaches the destination node by exploring the neighbouring nodes in each layer. While doing this, he colours the odd number of nodes with red color and even number of nodes with blue color. a. What algorithm can be chosen for this? b. What type of graph can you infer from it? Explain. 	CO2
6.	a) Given $f(n) = 8n^2 + 3n + 3$, Write a program to prove that $f(n) = O(n^2)$ and find the n_0 value. Plot a graph for $f(n)$ and $c * g(n)$ where c is a constant and for varying n values (10 to 30). b) A truck driver is given a set of locations to be covered with their distances by a company. The company strictly orders that truck should be started from a particular location. Design and implement an algorithm that gives a greedy solution to the truck driver's problem and display the shortest path for a given source location to all other locations. State the design strategy used and time complexity of the same.	CO3

